Organizing Code for Simulation Studies

Zivich Research Group

Simulation Studies

Simulation studies are a key aspect of methods work

- Verify expected statistical behaviors
- Illustrate important concepts
- Compare finite-sample performance between competing methods
- Check whether code works as intended
- Perform power calculations

Scenario should guide choices like: data generating mechanism, performance metrics, etc.

- Once those are decided, consider implementation

Code Organization

An organization to consider

```
Project/
    Code/
        output/
        data_generating_mechanism.file
        estimators.file
        compute_truth.file
        run_simulation_scenarios.file
        analyze_scenarios.file
        helper.file
```

Philosophy: each aspect lives in own file, so any changes carry forward everywhere

Data Generating Mechanism

Script contains all functionality to generate simulated data

- Functions to generate truth and observed data
- One overall function to generate complete data set
  - Basic argument to include: n, sample size
  - Optional: scenario when have many options
- Other functions can be included, but only the overall constructor should be imported elsewhere

Estimators

Script containing all estimators under consideration

- Separate functions for each estimator
- Generic inputs for future flexibility

Helper functions

- Script that contains any helper functions you might need for a simulation

Running Simulations

Compute the truth

- Own script that computes the truth and saves it as a reference (like in `output`)

Simulation scenarios

- Exact organization depends on run-time
- Best to have a general script
  - Script has a key variable which determines the scenario
- Only calls in objects from other files
- Saves all simulation outputs to `output/`

Analyze Outputs

Separate script that

- Loads in results from `output`
- Creates all summary tables and figures

Example

A basic (bad) example:
`github.com/pzivich/publications-code/tree/master/ClusteringEE/simulations`

A basic (good) example:
`github.com/pzivich/publications-code/tree/master/PooledLogitEE/sims`

A complex (good) example:
`github.com/pzivich/publications-code/tree/master/CausalRNG`

Questions?